

Detecting exoplanets XO-1 b and HAT-P-36 b transiting with ARCSAT

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Abstract

Exoplanets XO-1 b and HAT-P-36 b were attempted to be detected via the transit method. Photometry of their parent stars was performed using observational data from two nights of 60 s exposures using the Sloan z filter with ARCSAT at Apache Point Observatory. Poor weather limited the data quality, and the light curves, plotted using the `batman` Python package, of the parent stars proved inconclusive in detecting the exoplanets. The CCD gain was calculated as 1.4 electrons/ADU, in agreement with the published value. The readout noise of the CCD was calculated to be 16.9 electrons, which is higher than the reported value likely due to poor conditions.

1 Introduction

Apache Point Observatory (APO) is located in Sunspot, NM, USA, and is home to the Astrophysical Research Consortium Small Aperture Telescope (ARCSAT): a 0.5 m Classical Cassegrain telescope¹. ARCSAT has a primary mirror of diameter 50 cm, focal ratio $f/8.0$ and equatorial mount. The camera used was the University of Washington Flare Camera (Flare-Cam)², with a field size of 1024×1024 pixels, and field of view $11.2' \times 11.2'$. The CCD is an APOGEE U-47UV $13 \mu\text{m}$ pixel, back-illuminated, UV enhanced CCD, with a readout time of 1 s for the whole field with 1×1 binning, and a pixel scale of $0.656''/\text{pixel}$. During the reduction process, the CCD was found to be characterised by a gain of 1.4 electrons/ADU and readout noise of 16.9 electrons (23.0 electrons when the weather was poor).

The two exoplanets that were attempted to be detected were XO-1 b (on 1 June) and HAT-

P-36 b (on 3 June). They are both gas giant exoplanets, orbiting G-type stars. Their masses are $0.83 R_J$ and $1.8482 R_J$; have orbital periods of 3.9 days and 1.3 days; and have orbital radii of 0.0488 AU and 0.0241 AU respectively [2, 3]. These were suitable targets due to their short transits on the order of 3 hours each. This meant during the observations it would have ideally been possible to observe the ingress and egress, with baselines before and after the transit.

2 Methods

2.1 Target

The primary objective was to observe XO-1 b: the first one discovered by the XO project [5]. Being a hot-Jupiter, i.e. a gas giant planet orbiting very close to its parent star [5], means the transit is more readily observable. With a depth of 18 and V-band magnitude of 11.3, XO-1 b is a very sensible target to study. It was high in the sky on the day of observation (maximum elevation 80° , shown in Fig. 1) with all of the transit visible throughout the night. This makes it optimal for viewing, due to minimising the seeing caused by the atmosphere.

The secondary target, HAT-P-36 b, was chosen due to observing time limitations. Using an exoplanet transit finder³, HAT-P-36 b was returned as a result with sensible parameters for observation.

2.2 Observations

2.2.1 Calibrations

For each exoplanet transit, five bias, dark and dome flat frames were each taken before observations. These were used for the CCD reduction process, where each set were averaged via taking the median after running a sigma clipping

¹apo.nmsu.edu/Telescopes/ARCSAT/Telescope/index.html

²apo.nmsu.edu/Telescopes/ARCSAT/Instruments/arcsat_instruments.html

³astro.swarthmore.edu/transits/

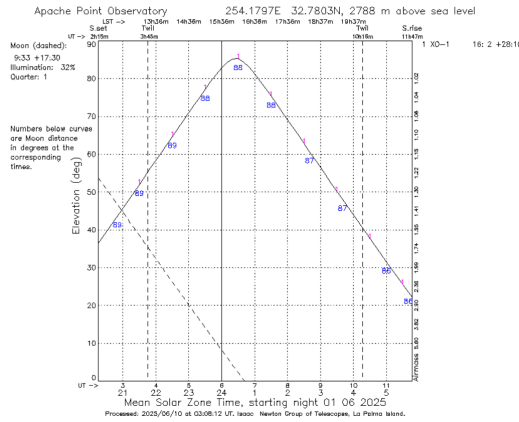


Figure 1: STARALT plot for XO-1 on 1 June (generated from STARALT at astro.ing.iac.es/staralt/)

algorithm on each frame. The code for the CCD reduction process can be found on GitHub⁴.

2.2.2 Filter

In deciding to use the Sloan z filter, the aim was to maximise the apparent magnitude of the star. XO-1 is brighter in the near infrared, which would make a percentage drop in flux due to the transit more visible. In addition, the aim was to minimise the limb darkening during ingress and egress. This limb darkening is caused by an apparent dimming of the star nearer to its edge, which is minimised in infrared wavelengths. In terms of the light curve, the transit depth is flatter and steeper, and so the decrease in flux will be clearer.

2.2.3 Science Frames

All science frames taken were using the z filter, and had exposure times of 60 s. Over the course of the nights, this led to 79 frames for XO-1 b and 104 frames for HAT-P-36 b.

2.3 Weather

The weather on all observing days was challenging. The first night (28 May), the dome was not opened at all so no data was acquired. On the second night (1 June), the sky was clear for the first half of the observing time. However, moisture was approaching and lightning was spotted in the distance around midnight local time and so the dome was closed. For the third night (3 June), there was light cloud throughout the

night, and thicker clouds around 00:30 local time meant the dome was closed. In addition, the illumination from the Moon was non-negligible for 3 June which impacted the data analysis.

2.4 Technical Issues

Few technical issues were encountered; the main issue was inclement weather. Before the observations on 3 June, there was an issue at 22:21 local time: *‘problem initialising the support library’*. This delayed observations so a smaller baseline was obtained, but luckily the system just needed to be rebooted by the 3.5 m observers on-site at APO.

2.5 Reduction

The data were reduced and analysed using the repository on GitHub⁴. The first step in the reduction process was to obtain median calibration frames. This was done using the `astropy.stats.sigma_clip()` function, using a $3\text{-}\sigma$ clipping about the median, and then a median was taken of these sigma-clipped frames. The darks were bias-subtracted, and the flats were bias- and dark-subtracted. The science frames were corrected by subtracting the bias and dark medians and dividing by the flat median. Cosmic rays were removed using the `astroscrappy.detect_cosmics()` function.

2.6 Photometry

Photometry was then performed on each reduced science frame on the target. This was done using the `photutils.aperture` submodule, and the best radius for the apertures was chosen by plotting radial profiles and chosen by eye when the increase in flux for a given radial increase is minimal. For these observations, this meant an aperture radius of 12 pixels. The centre of the apertures was found by initially making an estimate by opening the frames with DS9, and then using the `photutils.centroids.centroid_1dg()` function by fitting a 1D-Gaussian to the point spread function (PSF) to calculate the peak pixel coordinates. The photometry code was then run on all the frames.

⁴github.com/jbostock03/uw-astr480-arcsat

3 Results

3.1 Gain and Readout Noise

The variance of the difference between the two images, σ_{Δ}^2 , can be shown as,

$$\sigma_{\Delta}^2 = 2 \left(\frac{S}{G} + \frac{\sigma_{\text{RN}}^2}{G^2} \right)$$

where S is the signal level in ADU from one of the images, G is the gain in electrons per ADU, and σ_{RN} is the readout noise in ADUs [4]. Taking two images with high signal and below saturation, such as flats, this becomes,

$$G = 2 \frac{S}{\sigma_{\Delta}^2}$$

due to the readout noise, σ_{RN} , being very small. Or, taking two images with very low signal, such as biases, the equation can be reduced to,

$$\sigma_{\text{RN}} = \sqrt{\frac{\sigma_{\Delta}^2 G^2}{2}}$$

due to signal being zero by the definition of biases. The CCD was found to be characterised by a gain of 1.4 electrons/ADU and readout noise of 16.9 electrons for the XO-1 b observations (23.0 electrons for the HAT-P-36 b observations).

3.2 Light Curves

Light curves for the observations for XO-1 b and HAT-P-36 b were plotted, and are shown in Figs. 2 and 3 respectively. The light curve fits were calculated using the *batman* package [1] and inputting known parameters of the exoplanet systems [2, 3]. Due to poor weather, from the reduction process most frames had low fluxes and so each plot was zoomed in on for clarity. The full versions are available for viewing on the repository⁴.

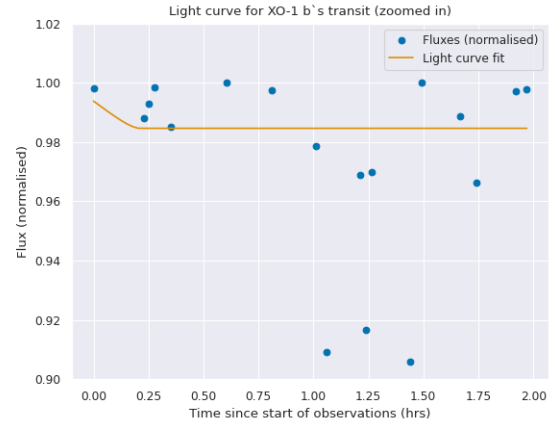


Figure 2: Light curve for the transit of XO-1 b on 1 June, starting at 22:23 local time.

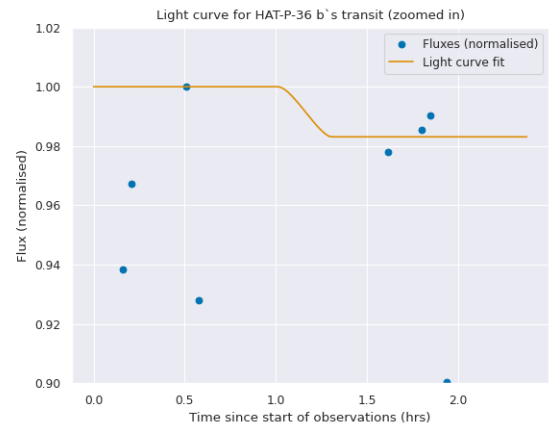


Figure 3: Light curve for the transit of HAT-P-36 b on 3 June, starting at 22:08 local time.

4 Discussion

From the *ARCSAT Instruments* page², the CCD gain is 1.25 electrons/ADU and the readout noise is 11.8 electrons. The gain is in good agreement with what was calculated from the observations, however, the readout noise is overestimated from the observations. Defects in the CCD may be affecting the estimations, which could be improved upon by using more images. Another contributor seems to be the gain. If the measured readout noise, 12.0 ADU is multiplied by the gain provided online, 1.25 electrons/ADU, a readout noise of 15.0 electrons is obtained, which is closer to the expected value. This is

still greater than the reported value so other contributions to the readout noise are unaccounted for.

The light curves for the transits are incomplete, due to the changing conditions the fluxes are inconsistent and so the detection of any transits are inconclusive. The transit of HAT-P-36 b is possibly more explainable using the four data points within 3% of the maximum flux detected. For a more decisive detection, the weather would have to be clear, and observations would take place over a long period of time; over multiple transits would be optimal.

5 Conclusions

In conclusion, the detection of exoplanets XO-1 b and HAT-P-36 b were inconclusive; follow-up observations over extended periods of time in clear weather are needed. The gain and readout noise of the FlareCam CCD used with ARCSAT were nevertheless calculated using the observational data.

Further exoplanet detection using ARCSAT remains hopeful. With inconclusive data from these two short observing periods, further observations of HAT-P-36 b may prove successful. In addition, spectroscopic observations of the system could lead to further understanding of the planet's composition and improving data on hot-Jupiters in distant star systems.

Acknowledgement

Based on observations obtained with Apache Point Observatory's 0.5-m Astrophysical Research Consortium Small Aperture Telescope.

References

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